Farhan Naeem

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Education

McMaster University Hamilton, ON
Bachelor of Applied Science in Computer Science – GPA 3.9/4.0 *Sept. 2025 – Exp. May 2029*
Coursework: Data Structures & Algorithms, Databases, Concurrent Systems, Computer Architecture, Algorithms & Software Design, Discrete Mathematics, Web Software Design

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL, C/C++, HTML/CSS, Java, PHP, Haskell, Elm
Frameworks and Libraries: React, Next.js, Tailwind CSS, FastAPI, Firebase SDK, NumPy, pandas, Matplotlib, PyTorch
Tools and Platforms: Git, Linux, GitHub Actions, Auth0, AWS, MATLAB

Experience

VP of Tech and Operations July 2026 – Present
McMaster Start Coding (NSERC-funded) *Hamilton, ON*

- Keep an **Elm** browser-based coding platform reliable for **11,000+** students across **491** classes in **6** countries by leading **8** developers through **200+** merged PRs while sustaining a **7-release-per-term** cadence
- Reduce a major class of beginner syntax errors by engineering **MacVenture 3.0**, a visual adventure-game builder and **code-generation pipeline** compiling **18** primitives into **120**-line programs, with **94%** of exports compiling on first try
- Cut shape-tool load time by **82%** (**1,050ms** to **190ms**) by reducing redundant model updates and batching renders, supporting **20-language** classrooms via **i18n JSON bundles** with community-contributed locale files

Software Engineer Intern June 2025 – Sept. 2025
PLASP *Mississauga, ON*

- Maintained uptime for Chromebooks, devices, and Wi-Fi serving **250+** students and staff by resolving **50+** hardware and network issues across **5+** school locations
- Reclaimed **20 hrs/week** of staff manual review and cut SIS data errors **15%** across **2K+** student records by replacing manual record checks with **Python** audit scripts that flag inconsistencies and auto-update the SIS, tracked over **3** monthly audit cycles
- Automated test-result analysis by building a **Python** data pipeline that ingests and persists **1,000+** data points per run, exports to **CSV**, and generates regression plots with **NumPy** and **Matplotlib**

Projects

EV Battery Health & Second-Life Tool – *Best Practical Solution, Purolator Hackathon | Python, HiGHS*

- Estimated health for **106** fleet EV batteries within **∼4%** error on unseen weeks by fitting a robust nonlinear model on **10K+** records, correcting for 10–23% charger losses and seasonal bias
- Predicted retirement dates via **200** block-bootstrap refits (2.4%/yr fade), finding **63** reusable packs
- Modeled **$17–19K/yr** savings per terminal with a 15-min **linear program** over **22.7K** charging sessions on real tariffs, exposing demand charges as 41–58% of the bill

AI Financial Planning Platform – *Hack Canada 2026, team of 4 | Next.js, FastAPI, Gemini, Backboard, React Flow*

- Removed manual refreshes from SMB planning with a **React Flow** canvas that triggers **2 parallel AI agents** **1.5s** after every edit
- Ranked **47+** Canadian cities on a **Mapbox** map by scoring **5 live data sources** (StatCan, FRED) in **FastAPI**
- Kept analysis available through AI outages and cut redundant API calls by pairing **Gemini** agents with persistent **Backboard** threads (auto-fallback) and **TTL caching**, plus **CSV/Google Sheets** ingestion into canvas nodes

Community Cleanup Tracking Platform – *Hack404 | Next.js, Flask, Firebase, Google Maps API, PyTorch*

- Logged **85+** verified community cleanups across Toronto by **40** members by building a full-stack platform with GPS-tagged photo submissions in **Next.js**, **Flask**, and **Firebase**
- Visualized city-wide litter hotspots on a real-time map with **Google Maps API** and **Cloudinary**
- Classified waste into **6** streams at **92%** accuracy across **500+** labeled cleanup photos by fine-tuning a **ResNet-50** backbone at **140ms** per image, selecting it over **YOLO** after benchmarking showed higher precision for single close-range objects